

Tarun Kumar Sahu

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EDUCATION

MIT Art, Design and Technology University

B.Tech in Computer Science and Engineering — CGPA: 8.55 / 10

Pune, Maharashtra

2025 – 2029

HACKATHONS & TECHNICAL EXPERIENCE

AI Tinkerers × Michelin Harness Engineering Hackathon | *Top 10*

2026

- Built **ReliAI**, a multi-agent industrial incident investigation system using specialized agents, adversarial validation, confidence assessment, and human-in-the-loop approval for more reliable AI-assisted analysis.

Bharatiya Antariksh Hackathon | *Team Leader, Team Zenith Lab*

2026

- Led the design of an **AI-Powered Climate Digital Twin** architecture for environmental analysis, visualization, and decision support after evaluating approaches from **50+ technical and research papers**.

PROJECTS

FreshFusion – Multimodal Fruit Freshness System | *CV, ESP32, FastAPI*

2026

- Built a multimodal fruit-inspection prototype integrating computer vision with ESP32-based MQ135 gas sensing and DHT11 temperature/humidity data; designed a FastAPI-based sensor-fusion workflow for AI-assisted freshness classification.

Jarvis OS – Agentic AI Research System | *Python, AI Agents, Automation*

2026

- Built a desktop AI assistant prototype with application/tool execution and automation workflows; currently extending it toward persistent memory, agent orchestration, and local/cloud model integration.

AI-Based Autonomous Weed Detection & Removal Robot | *YOLO, CV, ESP32*

2026

- Developing a YOLO-based crop/weed detection pipeline with crop-safety logic and designing a planned Raspberry Pi–ESP32 edge architecture for sensors, autonomous navigation, and mechanical weed removal.

Other Projects

- **AWR-Bot** – robotics and software-hardware integration; **Women's Safety Smart Wristband** – ESP32, GPS, IoT, and SOS alerts; **Penny Pilot** – React-based personal finance and budget tracking application.

ACHIEVEMENTS

Cyber Verse 2026 | *1st Prize, MIT-ADT University*

2026

- Won a technical competition featuring cybersecurity and blockchain quiz rounds along with DSA and coding challenges during Engineer's Day 2026.

Mathematics Olympiad | *Winner, MIT-ADT University – College Level*

2025

TECHNICAL SKILLS

Programming: Java, Python, C++, SQL, JavaScript

AI / ML: Machine Learning, Computer Vision, YOLO, Generative AI, Multi-Agent Systems

Backend / Web: FastAPI, REST APIs, React

Edge / IoT: ESP32, Raspberry Pi, Arduino, Sensors, GPS

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems

Tools: Git, GitHub, Docker, Linux

CERTIFICATIONS

IBM SkillsBuild – Foundations in Generative AI

2026

Tata Cybersecurity Analyst Job Simulation – Forage

2026

Deloitte Cyber Job Simulation – Forage

2025